

A Deep Learning Approach for Estimating SpO₂ Using a Smartphone Camera

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Abstract—The COVID-19 pandemic has highlighted the significance of both telemedicine and measuring physiological signals as peripheral oxygen saturation (SpO₂) at home. Aiming to increase the confiability of SpO₂ readings performed by smartphones, this study presents a novel deep neural network (DNN) model that predicts SpO₂ levels using fingertip videos captured by smartphones. The model was trained and evaluated using the leave-one-out method and compared against the SpO₂ values measured by a set of oximeters. The dataset used in this work is a differential of the study, unlike most similar studies which use a breath-holding protocol that presents only a few samples of low SpO₂ values; our tests were evaluated using a publicly available dataset in which volunteers' SpO₂ levels were measured in a range from almost 100% to approximately 70% by manipulating the percentage of oxygen inhaled by the volunteers, and the SpO₂ levels were kept on lower values for a considerable period of time. The proposed DNN model was compared against multiple variations in the ratio-of-ratios (RoR) method and the model proposed by the dataset creators. The results show that the DNN outputs have a strong relationship with the true SpO₂ measurements, as indicated by a Pearson coefficient (PC) of 0.95 and a concordance correlation coefficient (CCC) of 0.95, and also a small error, measured by a root mean squared error (RMSE) of 2.92% and a mean absolute error (MAE) of 1.97%. Compared with the RoR method, which achieved a PC of 0.29, CCC of 0.21, RMSE of 8.83% and MAE of 7.03%, the proposed DNN model exhibits superior performance. Moreover, the proposed DNN model outperforms the model proposed by the dataset authors, which achieved an MAE of 5.00%. These findings demonstrate that the proposed DNN model can accurately predict SpO₂ levels using smartphone-recorded fingertip videos with lower error and higher precision than a widely used classical method.

Index Terms—COVID-19, deep neural network (DNN), images, oxygen saturation, physiological signals, smartphone.

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I. INTRODUCTION

IN RECENT years, there has been a significant increase in the number of people using smartphones worldwide. With their increasing capabilities, smartphones are not just limited to basic communication and entertainment, but can also be used for multiple healthcare purposes, including the extraction of physiological signals. Smartphones are equipped with sensors such as accelerometers, microphone, and multiple cameras, and its hardware has been tested by researches for monitoring multiple physiological signals. Examples of physiological signals extracted by smartphones, or a similar hardware, are respiration rate (RR) [1], pulse rate (PR) [2], [3], [4], oxygen saturation (SpO₂) [3], [5], [6], [7], and blood pressure (BP) [3], [8].

Nemcova et al. [3], used smartphones to estimate PR, SpO₂, and BP. To estimate PR and SpO₂, the rear camera was used to capture videos of the user's fingertip, which were then processed to create a photoplethysmogram (PPG) signal. To estimate BP, the microphone was used to record the phonocardiogram (PCG), and BP was estimated based on the transit time from PCG and the camera's PPG. The authors achieved a mean absolute error (MAE) of 1.1% for SpO₂, 1.4 beats/min for PR, and in BP prediction, the systolic MAE was 5.1 mmHg and the diastolic was 7.5 mmHg.

In another study Bae et al. [9], evaluated techniques to infer the PR and RR of the user using smartphone cameras. Testing with 95 volunteers, they extracted the PR of the participants with a mean absolute percent error (MAPE) of $1.6\% \pm 4.3\%$. In the RR study, 50 volunteers participated, and the achieved MAE was 0.70 ± 0.67 breaths/min for healthy subjects and 0.80 ± 0.60 breaths/min for participants with chronic respiratory conditions.

Deep neural networks (DNNs) have been a promising tool to process biological data. In [10], convolutional models, long short-term memory (LSTM) models, and combination of both to detect sleep apnea were evaluated, and their results show that the combined model presented the highest accuracy (88.13%) to identify this disorder. In [11], a deep learning model was applied to estimate the BP using data extracted from face images, with the network fed with the remote PPG generated from the skin region and another seven features derived from the images. They were able to infer the BP with an MAE of 11.54 mmHg for systolic BP and 8.09 mmHg for diastolic BP. Lampier et al. [12] also applied a combination of convolutional and LSTM networks to infer the PR of people using face images. It was shown that for smaller signal sizes, the DNN outperforms a conventional method, achieving an

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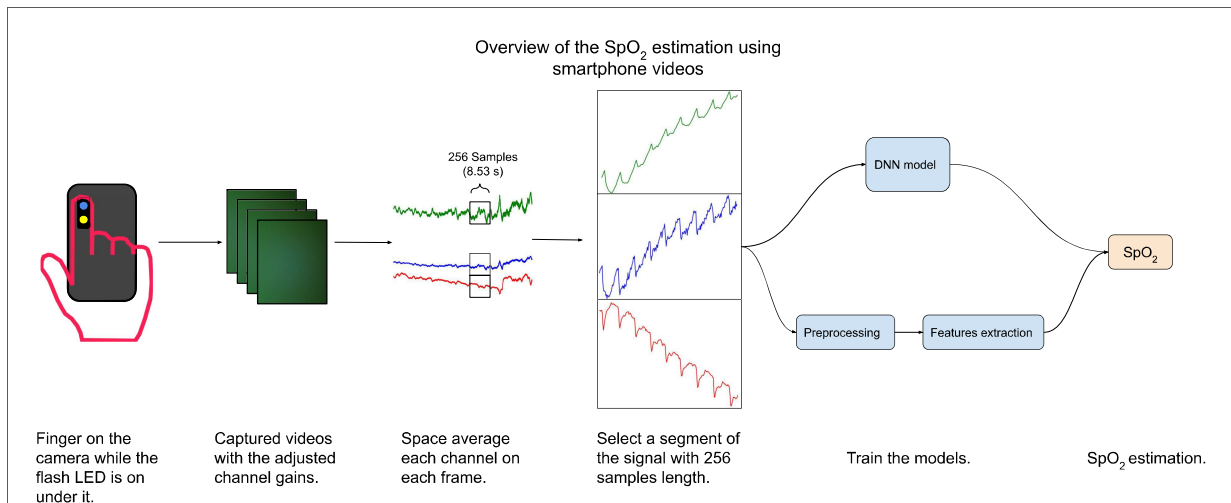


Fig. 1. Overview of the system used to infer SpO_2 levels.

MAE of 3.97 beats/min and root mean squared error (RMSE) of 6.47 beats/min when 5.12 s of data are used, and MAE of 3.00 beats/min and RMSE of 5.45 beats/min when 10.24 s of data are used.

During the COVID-19 pandemic, the peripheral blood oxygen saturation (SpO_2) level has received heightened attention in the scientific community, as it is a crucial indicator of respiratory function. The possibility to infer this physiological feature using the smartphone was also a research focus, as presented in [5], which evaluated the performance of three DNN models for retrieving SpO_2 measurements. The best performing model achieved an RMSE and MAE of 2.32%, as well as a correlation coefficient (CC) of 0.46. The models were evaluated on a dataset acquired using a breath-holding protocol, where the volunteers held their breath to decrease SpO_2 levels. It is worth commenting that collecting an SpO_2 dataset can be challenging, as healthy individuals typically have oxygen saturation levels above 95% and do not experience significant changes. To obtain a wider range of SpO_2 readings, some studies like [5] and [6] have used a breath-holding protocol. However, this method may not achieve a significant number of samples of SpO_2 levels below 90%.

This text presents the evaluation of a DNN model in the task of measuring SpO_2 values of a person using videos from their fingertip taken by a smartphone camera. Our tests were performed on a dataset acquired by clinically reducing and keeping the oxygen saturation levels from over 90% until levels near 70% for a considerable time. For the best of our knowledge, there is only one work [7] that performed similar tests, which is the same work that recorded and provided the dataset. We show that our model performed considerably better than the model presented in [7]. It is also presented that using only two color channels (red and green) provides a more precise estimation than using all three color channels.

This article is structured as follows: Section II details the dataset used, the architecture of the DNN used, and the train-test protocol. It also presents the implementation of a classical approach for estimating SpO_2 from videos, which

is used for comparison to the DNN. Section III presents and compares the error and correlation metrics of both our method and the classical approach. In Section IV, our results are compared with similar works in the literature, and a deep analysis about the results is performed. Finally, in Section V, we summarize our findings and explain the impact of our work on improving people's well-being.

II. METHOD

The methodology used in this work to predict SpO_2 using DNN and smartphone videos is presented in Fig. 1. The recordings used in this work were provided by the dataset from [7], in which the authors captured videos of people's fingertip while the camera's LED was on. Then the frames were processed by space averaging each color channel of the frames, and generating three time series, one for the red channel, one for the green and one for the blue. We then segmented the signal into windows of 256 samples, approximately 8.53 s at 30 frames/s and feed the DNN to train and then predict the SpO_2 values. The training and testing steps were performed with different volunteers.

A. Dataset

In the study conducted in [7], six subjects inhaled multiple concentrations of oxygen–nitrogen medical mixture in a controlled hospital setting for 14–19 min. The oxygen concentration of the mixture started at 18%–21% and was decreased in steps every 1–2 min aiming to change and keep for 30 s the oxygen saturation of the subjects in four different ranges: above 93% (room air), 89%–93%, 80%–88%, and 70%–79%.

To capture video signals, two Google Nexus 6P smartphones were attached to both the index fingers of each subject, in which the fingers covered the whole camera. The smartphones were used to record fingertip videos at a frame rate of 30 frames/s, and with the flash LED on. A custom application was developed for the purpose of configuring the camera settings and recording the videos. The camera exposure

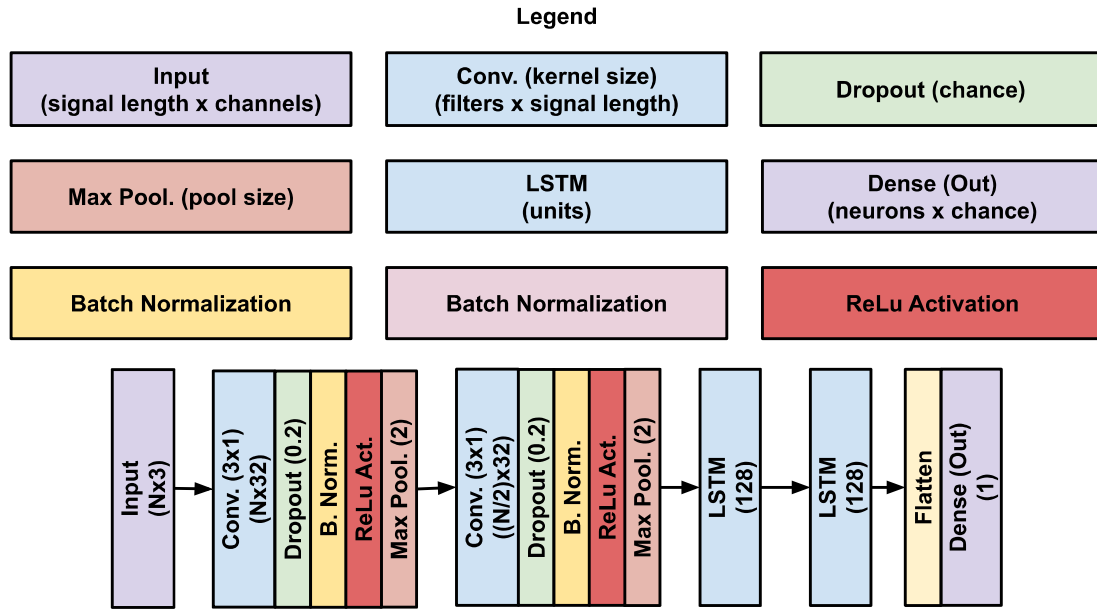


Fig. 2. Deep learning model architecture.

settings remained constant throughout the experiment, and the color gains were set to $1\times$ for the red channel, $3\times$ for the green channel, and $18\times$ for the blue channel, as determined empirically by the authors. Each recording session lasted between 14–19 min, and during the recording process, the flashlight of the smartphone was kept on while ice clay packs were used under the device to prevent overheating. It was also ensured that the packs had no contact with the hand. Along the tests, the ambient light conditions were maintained constant in a hospital-like room with fluorescent white light.

The SpO₂ was recorded using multiple oximeters, including the *Masimo Radical-7*, which was been selected by the authors as the ground truth. The mean difference between the reference *Masimo Radical-7* and the other oximeters was less than 1%; however, the limits of agreement were varied widely, which, according to the authors, highlight the differences between approved pulse oximeters.

The authors provided the smartphone data in the form of raw videos and RGB signals, which were extracted by space-averaging each color channel in the frames, resulting in $3 \times N$ signals (three color channels and N frames). In our work, we used the RGB files of the right hand provided by the authors as input to our DNN model, and the SpO₂ reference values were obtained as the median of the five oximeters attached to each volunteer.

B. Deep Learning Model

The video captured by the smartphone is preprocessed, which involves the generation of an RGB time series with dimensions of $3 \times N$ (the three color channels and the N frames). A segment of 256 samples of this processed data is then fed as input to the DNN, which yields the predicted value of SpO₂ as the output.

To process the smartphone data, a model consisting of convolutional and LSTM layers is proposed. Here, this approach

merges the feature extraction capabilities of convolutional layers [13], while also taking advantage of the time-dependent nature of signals captured by the LSTM layers [14]. Unlike the model proposed by [7], which only uses convolutional layers, our model begins with two sets of convolutional layers, each containing 64 filters of size 21 and activated with the rectified linear unit (ReLU) function. After each convolutional layer, a batch normalization and a maxpooling layer with size 4 are concatenated. Also, dropout layers with a probability of 10% are added to prevent overfitting. Following the convolutional sets, two LSTM layers with 128 units each, also activated with ReLU, are included, culminating in a single neuron with linear activation. The architecture is presented in Fig. 2.

The leave-one-out method was used to train the model, where five out of six volunteers were used for training, and the remaining volunteer was held out for testing. This process was repeated six times to ensure that each volunteer was included in the test group at least once. Multiple color channels were tested at the DNN input: all three color channels, combinations of two color channels, and only a single channel. The length of the input was set to 256 samples, which is equivalent to approximately 8.53 s of data.

To reduce overfitting, a data generator was implemented to randomly select a portion of the signal and segment it while training, instead of splitting the signal in prefixed points. The models were trained for 70 epochs, and the DNN model was evaluated by computing SpO₂ at intervals of 128 samples (equivalent to 4.27 s).

C. Ratio-of-Ratios Method

To directly compare our method, a classical approach was also implemented, the ratio-of-ratios (RoR) method [15], which was originally developed for use in pulse oximeters and subsequently adapted for imaging SpO₂. The operating principle of oximeters, both conventional and video-based,

is based on the differential absorption spectra of oxyhemoglobin (OHb) and deoxyhemoglobin (DHb), particularly in the red and infrared regions [16]. In medical pulse oximeters, the SpO₂ levels are estimated by projecting red and infrared lights onto the skin and determining the relative proportions of each wavelength that pass through the blood vessels or are reflected by them, depending on the device model. The signals detected by the oximeter's red and infrared sensors are PPGs, and the RoR method described in [15] can be used to calculate SpO₂ from the red and infrared PPGs. This involves estimating the alternating (ac) and direct (dc) components of the PPG signals, applying (1) to obtain an R value for both the red and infrared PPGs, and then using (2) to derive the R_{RoR} value. A linear regression is then performed on a portion of the data to calibrate the system, and then SpO₂ can be retrieved from new data

$$R_{\text{ch}} = \frac{\text{ac}}{\text{dc}} \quad (1)$$

$$R_{\text{RoR}} = \frac{R_{\text{red}}}{R_{\text{infrared}}}. \quad (2)$$

Various adaptations of the RoR method for use with videos have been proposed in the literature. However, since smartphone cameras cannot capture the infrared spectra, only the visible one, some authors have used combinations of the red channel with either the blue channel [6] or the green channel [17], while others have used all the three channels [18] to adapt the RoR method. To provide a comprehensive comparison, we conducted the RoR method with all possible combinations of two color channels, as well as using all the three colors. To use all the three colors of the video, we relied on the method proposed in [18], which used the features presented in (3). The ac features were extracted by computing the standard deviation (SD) and the peak-valley difference (PVD) of the PPG signals. The PPG signals from the red, green, and blue channels were preprocessed using a fourth-order bandpass Butterworth filter with cutoff frequencies of 0.5 and 4 Hz. The most heavily noisy parts of the signal were also removed by excluding segments of the series in which the difference between peaks was longer than 1.5 s, which corresponds to 40 beats/min, indicating that the signal was too noisy for a peak to be detected during that interval. After feature extraction, the RoR methods were calibrated using linear regression

$$R_{\text{RoR-RGB}} = \frac{R_{\text{red}}}{R_{\text{green}}}, \frac{R_{\text{red}}}{R_{\text{blue}}}, \frac{R_{\text{blue}}}{R_{\text{green}}}, R_{\text{red}}, R_{\text{green}}, R_{\text{blue}}. \quad (3)$$

To test the RoR method in multiple approaches, the ac component was obtained using two techniques: SD or PVD of the PPGs cycle. The dc was extracted by calculating the mean value of the PPG inside the window. And all the combinations of two color channels were tested using (2), and also the three color channels together were tested using (3). To evaluate the RoR method, the same evaluation points of the DNN were taken: a 256 samples' sliding window, with a step of 128 samples, was applied, generating a total of 1201 samples from all the six volunteers.

TABLE I
COMPARING THE DNN METRICS WITH VARIANTS OF RoR

Method	Color Channels	RMSE (%)	MAE (%)	MD $\pm$ SD (%)	CC	CCC
RoR-SD	RG	10.46	8.19	1.70 $\pm$ 10.43	-0.11	-0.08
	RB	10.17	8.24	1.73 $\pm$ 10.15	-0.19	-0.12
	BG	8.73	7.03	1.75 $\pm$ 8.72	0.29	0.21
	RGB	9.34	7.52	1.80 $\pm$ 9.33	0.28	0.25
RoR-PV	RG	10.71	8.22	2.01 $\pm$ 10.62	-0.03	-0.03
	RB	10.06	7.03	2.02 $\pm$ 10.00	-0.12	-0.03
	BG	8.73	7.03	1.75 $\pm$ 8.72	0.29	0.21
	RGB	9.73	7.78	2.38 $\pm$ 9.69	0.23	0.22
DNN	R	9.08	6.56	-1.39 $\pm$ 9.06	0.52	0.51
	G	4.77	3.24	0.77 $\pm$ 4.55	0.87	0.86
	B	5.39	4.13	0.27 $\pm$ 5.19	0.82	0.81
	RG	2.92	1.97	-0.12 $\pm$ 2.92	0.95	0.95
	RB	4.10	2.95	2.95 $\pm$ 3.97	0.90	0.88
	BG	8.34	4.31	2.38 $\pm$ 9.69	0.66	0.64
	RGB	4.63	3.83	1.81 $\pm$ 4.57	0.86	0.86

III. RESULTS

The results of SpO₂ prediction using the RoR and DNN methods are presented in Table I. We tested all the combinations of two colors, as well as a combination of the three color channels for both, DNN and RoR.

To evaluate the performance of the proposed architecture, we computed several commonly used regression metrics: RMSE, MAE, median (MDN) $\pm$ SD, CC, and concordance CC (CCC). The RMSE, MAE, and MDN $\pm$ SD metrics quantify the difference between the predicted SpO₂ values and the true values. The CC and CCC metrics measure the correlation between the predicted and true values, with values ranging from -1 to 1 . A value of -1 represents a perfect inverse correlation, a value of 1 represents a perfect direct correlation, and a value of 0 indicates no correlation. The difference between CC and CCC is that the former is insensitive to the mean error, while the latter is affected by the mean error.

It is notable in Table I that overall, the DNN outperformed the RoR method. The DNN presented relatively poor metrics only when only the red channel alone or the blue and green together are used, which is interesting because the blue-green combination resulted in the best RoR metrics.¹ It is also notable that the model presented better metrics when two channels are used instead of three. The red and green combination (RG-DNN) resulted in best results. The model trained with the three channels, RGB-DNN, had worse metrics than the RG-DNN and the model trained with the red and blue channels, RB-DNN. The DNN configurations that presented metrics as worse as the RoR methods were the GB-DNN and the R-DNN. Even G-DNN and B-DNN, which used a single channel, presented better results. As the RG-DNN presented clearly better results than the other methods, we performed a deeper analysis on it by estimating the error of all the subjects individually.

The metrics of the RG-DNN for each volunteer are presented in Table II. Notably, five out of six volunteers had RMSE values under 3% and MAE values under 2.25%. In addition, the all CCs were very close to 1, indicating a strong relationship between the predicted and reference SpO₂ values. Only subject 6 presented suboptimal results.

¹To make the identification of the DNNs easier, their setup will be identified in this text as <CHANNEL>-DNN, for example, the DNN trained with the red and blue channels will be RB-DNN.

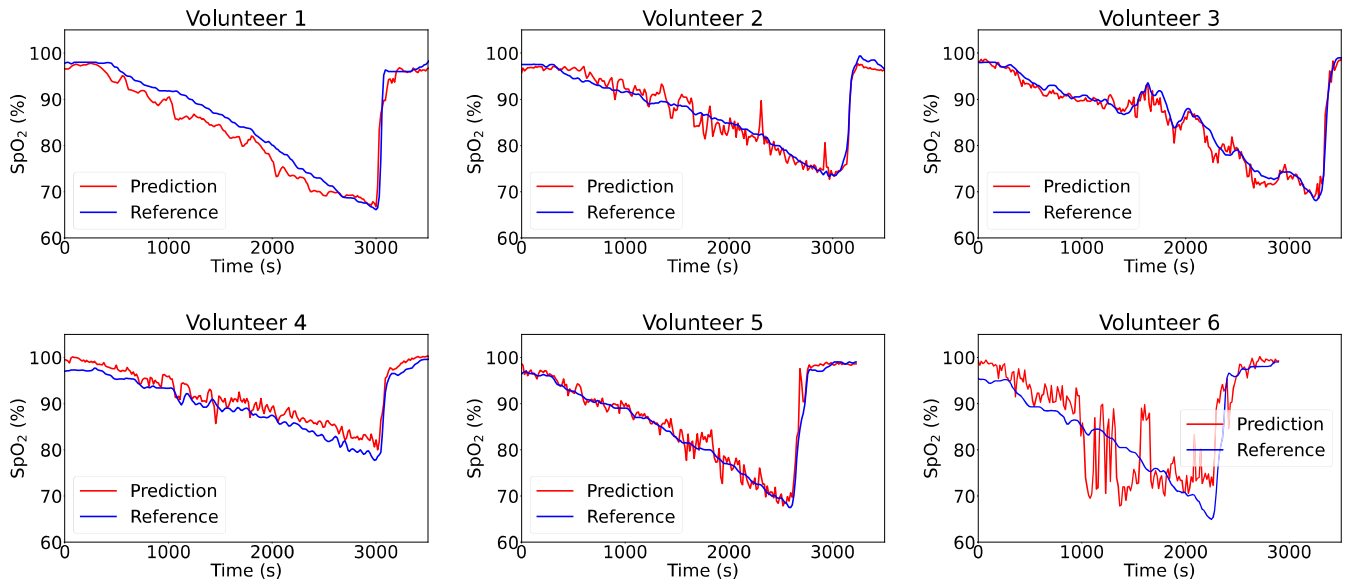


Fig. 3. SpO₂ predicted (red) and reference (blue) for each volunteer using the RG-DNN model.

TABLE II
RESULTS OF THE DNN FOR ALL SUBJECTS

Subject	RMSE (%)	MAE (%)	MDN $\pm$ SD (%)	CC	CCC
1	2.80	2.23	2.08 $\pm$ 2.41	0.96	0.98
2	1.61	1.25	0.24 $\pm$ 1.61	0.98	0.98
3	1.58	1.17	0.58 $\pm$ 1.47	0.99	0.98
4	2.25	1.99	-1.87 $\pm$ 1.18	0.98	0.93
5	1.76	1.15	-0.39 $\pm$ 1.67	0.98	0.98
6	5.96	4.54	-1.85 $\pm$ 5.80	0.84	0.83

TABLE III
HYPOXEMIA CLASSIFICATION

Volunteer	Accuracy	Precision	Recall	<i>F1</i> -score
1	0.90	0.83	1.00	0.91
2	0.92	0.99	0.86	0.92
3	0.89	0.86	0.94	0.90
4	0.87	0.99	0.75	0.85
5	0.96	0.98	0.95	0.96
6	0.82	0.99	0.74	0.85
All	0.89	0.93	0.88	0.90

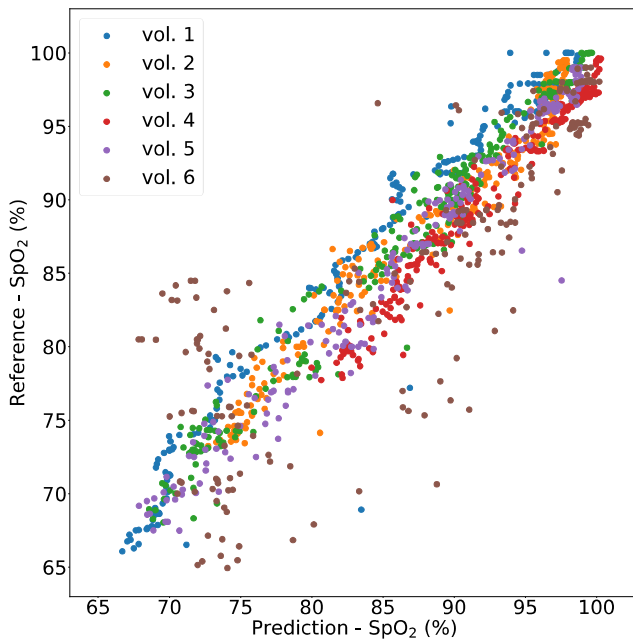


Fig. 4. Scatter plot of the prediction of all the volunteers using the RG-DNN model.

Fig. 3 presents the individual prediction of each volunteer performed by the RG-DNN model. Also, a general scatter plot with all the predictions is presented in Fig. 4.

We also calculated the performance of the RG-DNN model in classifying SpO₂ levels under 90%, a threshold commonly

used to define hypoxemia [19]. The precision of our model in identifying such samples was found to be 93%, while the total *f1* score and accuracy were 0.90% and 89%, respectively. These results suggest that our model exhibits high precision in detecting hypoxemia. The individual classification results for all the volunteers are presented on Table III. It is notable that all the volunteers then had an accuracy over 85%.

IV. DISCUSSION

Examining the results in Table I, it is clear that the overall performance of the DNN models surpasses the RoR techniques. The correlations of the RoR techniques are closer to 0, having only significant values observed when using the blue and green channels, or all the three channels, which resulted in CC values between 0.23 and 0.29.

It is interesting that unlike the majority of RoR applications reported in the literature, the RoR method presented the best metrics when the green and blue channels are applied. This finding could be attributed to the modification of camera gains for each channel during dataset generation. Specifically, the green channel was multiplied by a factor of 3, and the blue channel by a factor of 18, while the red channel had a gain of 1. However, when the green and blue channels were used to train the model, it had a high error, so it is not possible to affirm whether these channels have the best information for SpO₂ overall.

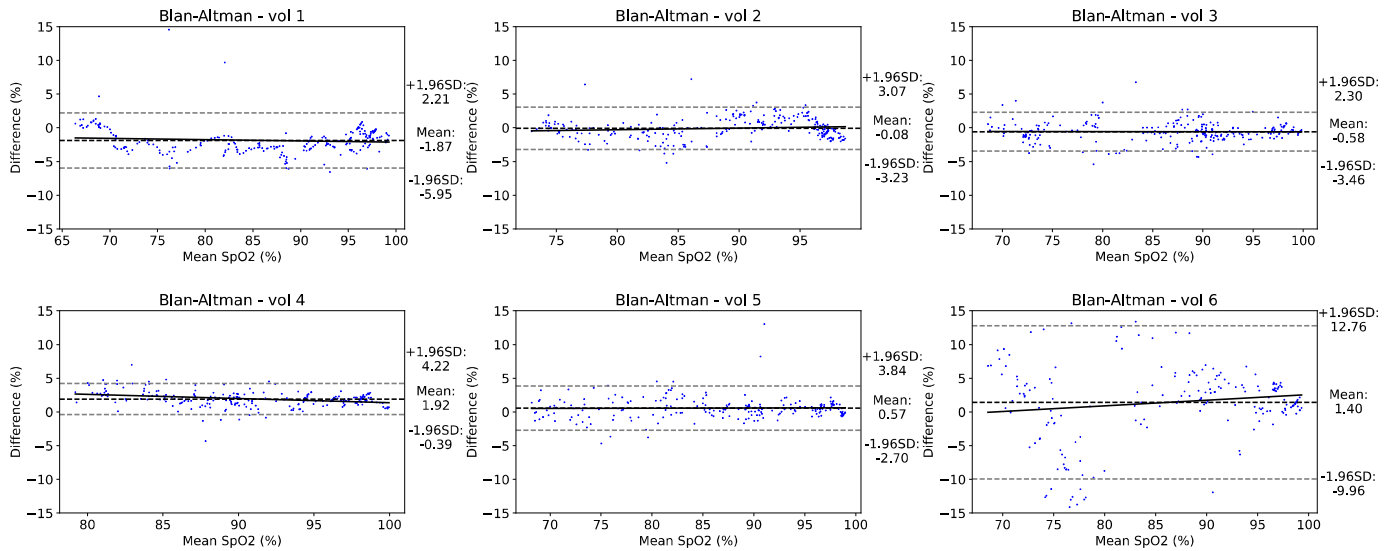


Fig. 5. Bland–Altman plots of the prediction from each subject using the RG-DNN model.

The superior performance of the DNN models is attributed to the DNN's ability to optimize filtering, feature extraction, and the regression tasks, extracting essential information from noisy data. Manual extraction of this information is much more challenging, especially since digital filters cannot separate noise inside the desired signal frequency spectra. Also, the use of a smartphone camera as the SpO₂ sensor introduces additional sources of noise, such as finger movement and changes in illumination, which the oximeter mitigates.

The results achieved by our DNNs, specially the RG-DNN, provide evidence that SpO₂ estimation using a smartphone camera is feasible with high precision. Overall, an RMSE lower than the maximum error limit recommended by the U.S. Food and Drug Administration [20] was achieved, which is 3.5%, and our best RMSE was 2.92%, with the RG-DNN. From the evaluation metrics in Table II, it can be seen that this setup measured SpO₂ levels of volunteers 1–5 with an RMSE of less than 3%, MAE lesser than 2.23%, low median error and SD, and correlation and concordance metrics over 0.93. Only volunteer 6 presented a higher error. It can be seen in Fig. 3 that the higher error is presented in regions between 1000 and 1800 s of measurement approximately, where error higher than 10% in that interval was obtained. This may be due to a small finger movement or even a change in pressure between the finger and the camera lens.

As the RG-DNN presented the best results, a Bland–Altman analysis was performed on the outputs of each subject for this setup. Fig. 5 displays the Bland–Altman plots of all the subjects. As demonstrated in Fig. 5, with the exception of volunteer 6, the outer lines (± 1.96 SD) were relatively thin for most volunteers, with less than 10% of distance between them, indicating a relatively low random error. Nevertheless, the low values of the mean error also suggest a low systematic error. In addition, for the RG-DNN, the correlation between the predicted and true SpO₂ was found to be very high. Examination of the CCs presented in Table II reveals individual correlations

above 0.84, and Table I presents that a correlation of 0.95 was obtained for predictions of all the volunteers combined.

The results of RG-DNN can also be compared with the metrics found in [7], in which a DNN based on convolutional layers was also used to estimate SpO₂, and an MAE of 5.00% was reported. Our best model achieved a smaller MAE error of 1.97%, lesser than half, indicating that the LSTM layers applied to our model helped to better infer the SpO₂ measurements with more precision.

It is also shown in Table III that the RG-DNN model is very precise in identifying hypoxia. The overall accuracy for the volunteers was 89%, where the lowest accuracy was for subject 4 which was 82%. The overall precision was 93% and the lowest was 83%, for subject 1. The overall recall was 88%, and the worst recall was estimated with subject 6, with 75%. All f1 scores were high, with all of them greater or equal 0.81, meaning a highly reliable model.

Comparing the results of the RG-DNN with other works that used different datasets, it is worth noting the differences between the results obtained using the breath-holding method, as performed in [5], and the dataset used in this study. Evaluating on a breath-holding method dataset, [5] achieved a CC of 0.46, which is lower than ours, indicating a stronger relation between the predicted and true values in our results. However, they achieved MAE and RMSE, 1.97% and 2.32%, respectively, which are closer to our results. A low error and also a low CC imply that the use of the breath-holding technique may not be optimal for evaluating video-based SpO₂ methods if one aims to achieve the same level of confidence as with a pulse oximeter, since the breath-holding method does not generate a significant number of samples under 90% as presented in [5], meaning that the correlation between the predictions and the measurements may not be well-measured. The approach of altering the ratio of inhaled oxygen generates a broader range of SpO₂ values, which has a direct impact on error metrics since greater variability in the

values may lead to a greater divergence between the predicted and actual values.

Regarding the processing time of the DNN, our model took 44.3 ms to process each sample, in a computer with Ryzen5 CPU and a RAM memory of 16 GB. The same setup was used to calculate the processing time of the RoR method, using peak detection to exclude outliers, which took 50.3 ms to process each sample. When only using the SD to calculate the features and without outlier filtering, it took 1.2 ms per sample. Besides being slower than the raw RoR method applied using SD, the DNN was faster than the RoR when peak detection and outlier rejection were applied.

V. CONCLUSION

In this study, we presented a novel DNN model composed of convolutional and LSTM layers for SpO₂ estimation using smartphone-camera-recorded videos of individuals' fingertips. To validate the efficacy of our model, we used a public dataset collected in a clinical setting where SpO₂ values ranged from approximately 70% to 100%. Our experimental results demonstrate that our proposed model outperforms a commonly used video-based SpO₂ estimation method. Furthermore, our method achieves a smaller error than a DNN architecture proposed by the dataset authors, with an RMSE of 2.92% and a CC of 0.95, indicating a high correlation between the predicted and actual values, and an overall error lower than the recommended by the FDA.

An interesting finding in this work is that the DNN trained only with the red and green channels performed better than the DNN trained with the three color channels. However, comparing with the RoR methods, the best metrics were achieved with the blue and green channels. Choosing the best channels for each method raises an interesting point for future studies.

Our research findings demonstrate that smartphone cameras can be used to estimate SpO₂ readings. In fact, to use smartphones for blood oxygen saturation measurement holds significant potential benefits for individuals, especially those with respiratory or cardiovascular conditions. By placing a finger over the camera lens and flash, a smartphone application can detect the ratio of OHb-DHb in the blood and display the resulting SpO₂ reading on the device screen. This technology has the potential to aid individuals in tracking their SpO₂ levels in real-time and monitoring changes, especially during physical activities or situations where oxygen levels may be compromised. Moreover, it can serve as an early warning system for those with chronic conditions, allowing them to take appropriate measures before symptoms worsen. The ability to measure SpO₂ using a smartphone offers individuals an opportunity to improve their health and well-being by taking greater control over their condition and potentially enhancing their quality of life.

Besides the promising results, there is still a wide range of real-world situations that are needed to be tested before clinical use: How does the model handle different skin tones? How does the finger pressure on the camera affect the measurements? What is the impact of the environment light in the readings? Do other physiological signals such as BP affect

the results? How much power and what is the response time of the model when embedded on a smartphone? What is the minimum hardware requirement of the smartphone to run it? This field of research can still be improved in the future, and as a suggestion, it is recommended to start a wider dataset captured on similar conditions, with subjects with multiple skin tones, measuring also other physiological signals, such as BP, and also changing the environment lighting.

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